TEXAS A&M UNIVERSITY Engineering

Solar Home Lighting

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Problem Definition

Current smart home lighting systems rely heavily on grid power and cloud connectivity, leading to delayed response times, reduced reliability, and loss of functionality during network outages. Existing solutions also fail to effectively integrate local control, sensor-based automation, and remote monitoring into a unified system, resulting in poor user experience and inefficient energy usage.

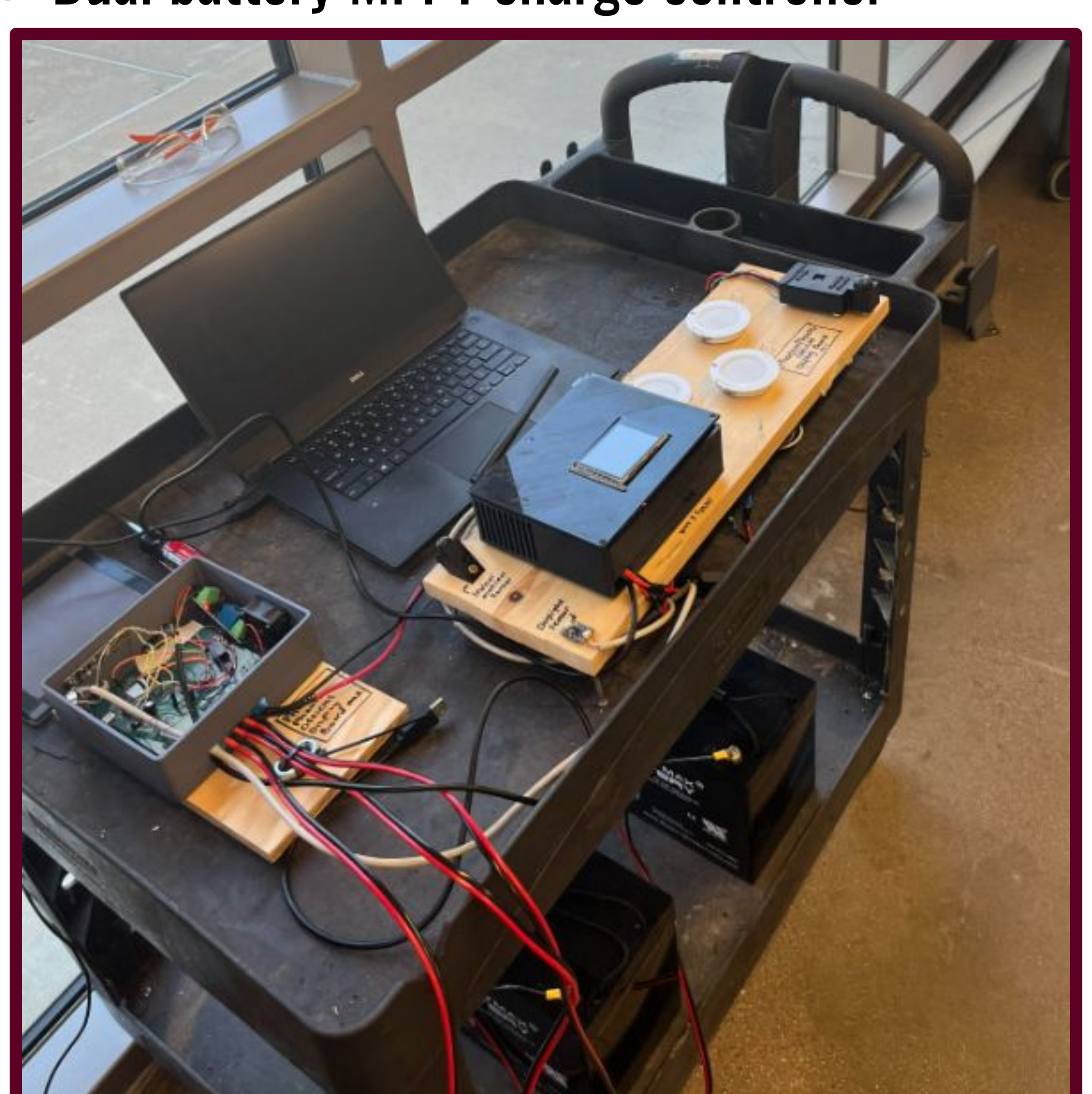
Methodology

Objective

- Develop a solar-powered lighting system with reliable local control and remote monitoring.

System Architecture:

- ESP32, FreeRTOS separates control, UI and network tasks
- Firbase used for remote synchronization
- Dual battery MPPT charge controller



Local Control:

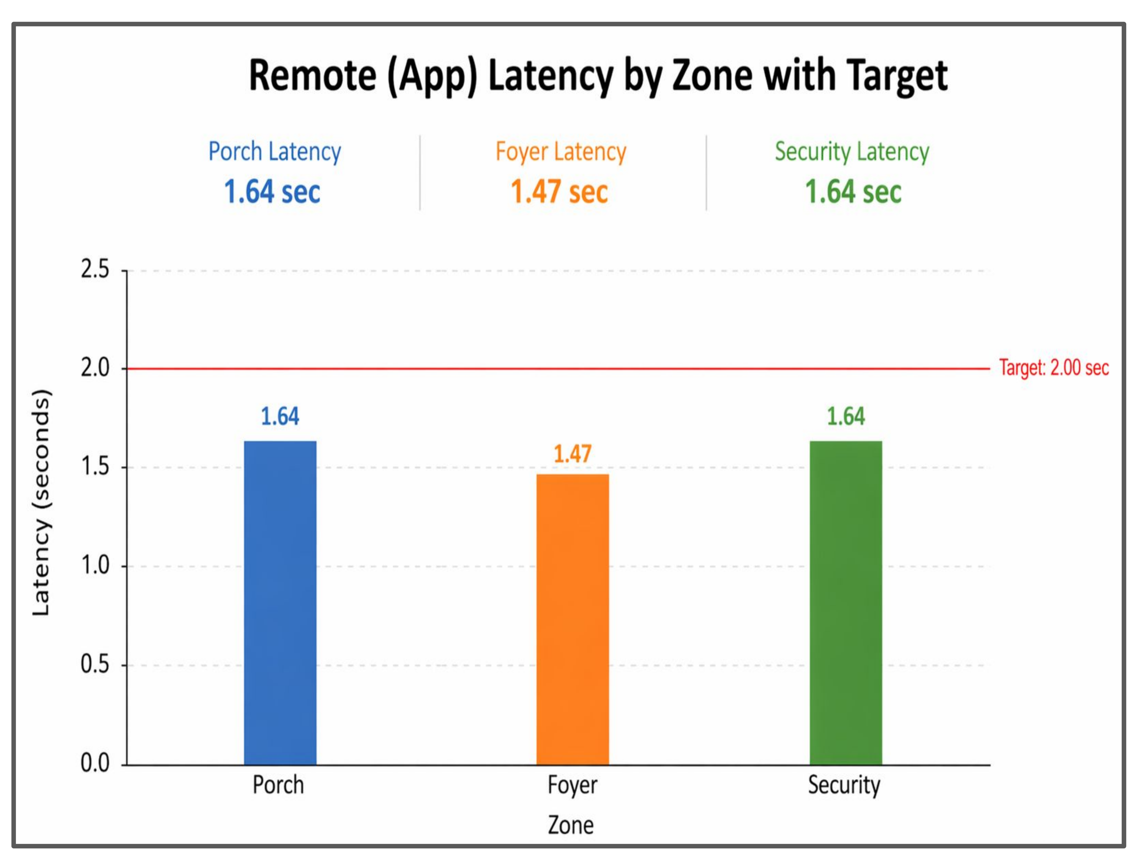
- 50 ms control loop enables real-time sensor automation
- Touchscreen UI with realtime system status and metrics

Remote Communication:

- App Communicates via Firebase
- System uploads state and executes commands asynchronously

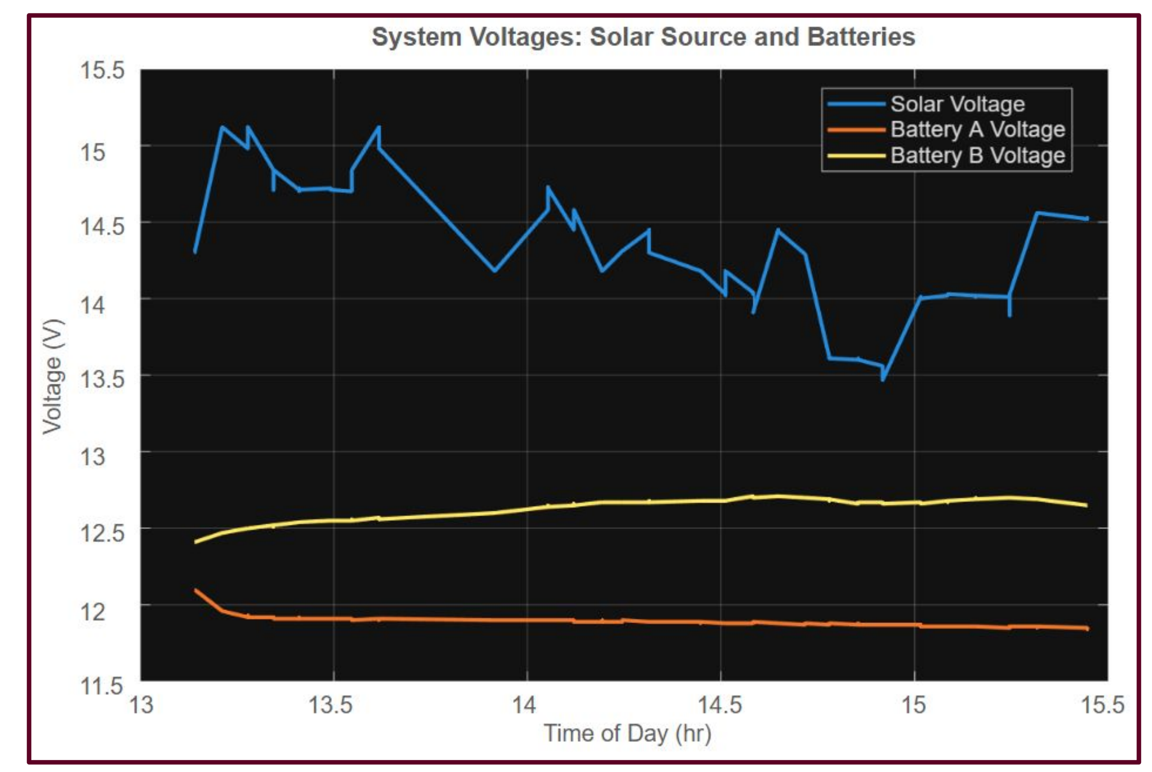
Project Results

Hybrid Control Performance: Achieved reliable operation across both local and remote interfaces, enabling immediate response for time-critical actions while maintaining full system accessibility and monitoring through the mobile application



Autonomous System Reliability: Demonstrated consistent sensor-driven response to environmental conditions, enabling dependable lighting and security operation without user intervention

Energy Management: Successfully implemented MPPT-based solar charge control with dynamic switching between dual batteries, enabling efficient energy harvesting, balanced battery utilization, and continuous system operation under varying environmental and load conditions



Discussion

System Robustness: Realtime local control ensures continued operation during network delays or outages, maintaining critical lighting functionality.

Design Tradeoffs: Balancing cloud synchronization with real time control increases system complexity but improves flexibility and accessibility.

Scalability Considerations Firebase based architecture support expansion to additional devices, sensors, and features with minimal changes

Conclusions

Energy Independence & Reliability:
Enables continued lighting and security operation during grid outages through solar powered dual battery energy storage, improving reliability in both residential and off-grid environments.

Improved Energy Efficiency & Sustainability: Decreases overall energy consumption by using sensor driven automation, and real time system response, minimizing unnecessary lighting usage.

Enhanced Quality of Life & System Resilience: Improves user experience by maintaining an operational system regardless of network connectivity, and power outages.

Increased Home Security & Awareness:
Strengthens residential safety by enabling immediate, automated responses to environmental events. Including motion triggered lighting and security notifications.